CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 4: Quadratic Equations

Topic-wise short notes with solved examples

A quadratic equation has the variable squared. This chapter covers how to recognise one, how to solve it by factorisation and by the formula, and how the discriminant tells the nature of its roots.

1 Standard form

- A **quadratic equation** in x is an equation of the form

$$ax^2 + bx + c = 0, \quad a \neq 0$$

- a, b and c are real numbers. a is the coefficient of x^2, b the coefficient of x and c the constant term.
- To check whether an equation is quadratic, simplify it and see whether the highest power of x is 2.
- A number α is a **root** (or solution) of the equation if $a\alpha^2 + b\alpha + c = 0$. A quadratic equation has at most two roots.

Example 1.1 Check whether each equation is quadratic: (i) $(x+1)^2 = 2(x-3)$ (ii) $x(x+3) = x^2 + 5$.

SOLUTION

(i) $x^2 + 2x + 1 = 2x - 6$, so $x^2 + 7 = 0$. The highest power is 2.

(ii) $x^2 + 3x = x^2 + 5$, so $3x - 5 = 0$. The x^2 terms cancel.

(i) is a quadratic equation. (ii) is not; it is linear.

2 Solving by factorisation

- Step 1: write the equation in standard form.
- Step 2: split the middle term bx into two terms whose **sum is** b and whose **product is** ac.
- Step 3: factorise by grouping.
- Step 4: put each factor equal to zero.

Example 2.1 Solve $x^2 - 5x + 6 = 0$.

SOLUTION

We need two numbers with sum -5 and product 6: they are -2 and -3.

$x^2 - 2x - 3x + 6 = 0$, so $x(x-2) - 3(x-2) = 0$ and $(x-2)(x-3) = 0$

$x = 2$ or $x = 3$

Example 2.2 Solve $3x^2 - 11x + 6 = 0$.

SOLUTION

$ac = 18$. Two numbers with sum -11 and product 18: -9 and -2.

$3x^2 - 9x - 2x + 6 = 0$, so $3x(x-3) - 2(x-3) = 0$ and $(3x-2)(x-3) = 0$

$x = \frac{2}{3}$ or $x = 3$

Example 2.3 Solve $2x^2 - 8x = 0$.

SOLUTION

Take the common factor: $2x(x-4) = 0$

$x = 0$ or $x = 4$

REMEMBER

Never divide both sides by x. In the last example that would lose the root $x = 0$.

3 The quadratic formula

- The roots of $ax^2 + bx + c = 0$ are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Use the formula when the equation does not factorise easily.
- Write the equation in standard form first, then read a, b and c with their signs.

Example 3.1 Solve $2x^2 - 7x + 4 = 0$ using the quadratic formula.

SOLUTION

$a = 2, b = -7, c = 4$

$b^2 - 4ac = 49 - 32 = 17$

$x = \frac{-(-7) \pm \sqrt{17}}{2 \times 2} = \frac{7 \pm \sqrt{17}}{4}$

$x = \frac{7 + \sqrt{17}}{4}$ or $x = \frac{7 - \sqrt{17}}{4}$

4 Nature of roots

- The **discriminant** of $ax^2 + bx + c = 0$ is $D = b^2 - 4ac$.

Discriminant	Nature of the roots	The roots are
$D > 0$	Two distinct real roots	$\frac{-b+\sqrt{D}}{2a}$ and $\frac{-b-\sqrt{D}}{2a}$
$D = 0$	Two equal real roots	$-\frac{b}{2a}$ and $-\frac{b}{2a}$
$D < 0$	No real roots	None

Example 4.1 Find the nature of the roots of (i) $2x^2 - 6x + 5 = 0$ (ii) $4x^2 - 12x + 9 = 0$.

SOLUTION

(i) $D = (-6)^2 - 4(2)(5) = 36 - 40 = -4 < 0$

(ii) $D = (-12)^2 - 4(4)(9) = 144 - 144 = 0$, and the roots are $-\frac{b}{2a} = \frac{12}{8} = \frac{3}{2}$

(i) No real roots. (ii) Two equal real roots, each $\frac{3}{2}$.

Example 4.2 Find the values of k for which $x^2 + kx + 16 = 0$ has two equal roots.

SOLUTION

For equal roots, $D = 0$: $k^2 - 4(1)(16) = 0$

$k^2 = 64$

$k = 8$ **or** $k = -8$

5 Word problems

- Let the unknown be x, form the equation from the given condition, solve it, and reject any root that does not fit the situation.
- A length, a speed, an age or a number of objects cannot be negative.
- For speed problems use $\text{time} = \frac{\text{distance}}{\text{speed}}$.

Example 5.1 The product of two consecutive even positive integers is 168. Find the integers.

SOLUTION

Let the integers be x and $x + 2$. Then $x(x + 2) = 168$, so $x^2 + 2x - 168 = 0$.

$(x + 14)(x - 12) = 0$, so $x = -14$ or $x = 12$. The integers are positive, so $x = 12$.

The integers are 12 and 14.

Example 5.2 The length of a rectangular plot is 4 m more than its breadth, and its area is 96 m². Find its dimensions.

SOLUTION

Let the breadth be x m. Then the length is $(x+4)$ m and $x(x+4)=96$.

$x^2+4x-96=0$, so $(x+12)(x-8)=0$. Rejecting $x=-12$, we get $x=8$.

Breadth 8 m and length 12 m.

Example 5.3 A train covers 300 km at a uniform speed. If the speed had been 10 km/h more, the journey would have taken 1 hour less. Find the speed of the train.

SOLUTION

Let the speed be x km/h. Then $\frac{300}{x}-\frac{300}{x+10}=1$.

$300(x+10)-300x=x(x+10)$, so $x^2+10x-3000=0$.

$(x+60)(x-50)=0$. Rejecting $x=-60$, we get $x=50$.

The speed of the train is 50 km/h.

6 Common mistakes

- Reading b or c without the sign. In $2x^2-7x+4=0$, $b=-7$.
- Dividing only part of the numerator by $2a$. The whole of $-b\pm\sqrt{D}$ is divided by $2a$.
- Using the formula before writing the equation in standard form.
- Keeping a negative root for a length, an age or a speed.
- Writing "no roots" when $D<0$. The correct statement is "no real roots".

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